Supplementary Information to Bridging the Gap in the Responsible AI Divides

A AI Safety Labs and People Used for Data Collection

We build on this AIS corpus by extending it with the metadata of AIS-related papers from a list of 35 labs and other individual authors chosen based on references from previous work [4, 8, 37, 93].

In addition, we included papers from the following organizations which represent work on AIS:

- Autonomous Agents Research Group (`https://agents-lab.org/`)
- Alignment of Complex Systems Research Group (`https://acsresearch.org/`)
- Anthropic (`https://www.anthropic.com/`)
- Association for Long Term Existence and Resilience (`https://alter.org.il/`)
- Apart Research (`https://apartresearch.com/`)
- Center for AI Safety (`https://safe.ai/`)
- Center for Human-Compatible AI (`https://humancompatible.ai/`)
- Center on Long-Term Risk (`https://longtermrisk.org/`)
- Center for the Study of Existential Risk (`https://www.cser.ac.uk/`)
- Cooperative AI Foundation (`https://www.cooperativeai.com/`)
- DeepMind (`https://deepmind.google/`)
- EleutherAI (`https://www.eleuther.ai/`)
- FAR.AI (`https://www.far.ai/`)
- Future of Humanity Institute (`https://www.futureofhumanityinstitute.org/`)
- Forecasting Research Institute (`https://forecastingresearch.org/`)
- Google Research (`https://research.google/`)
- Centre for the Governance of AI (`https://www.governance.ai/`)
- Global Priorities Institute (`https://globalprioritiesinstitute.org/`)
- Meta (`https://ai.meta.com/`)
- Model Evaluation & Threat Research (`https://metr.org/`)
- Mila (`https://mila.quebec/`)

- Machine Intelligence Research Institute (https://intelligence.org/)
- NeurIPS ML Safety 2022–2024 (https://neurips.cc/)
- OpenAI (https://openai.com/)
- Oxford Martin AI Governance Initiative (https://aigi.ox.ac.uk/)
- Palisade Research (https://palisaderesearch.org/)
- RAND (https://www.rand.org/)
- Redwood Research (https://www.redwoodresearch.org/)
- UCL Theory of Learning Lab (Saxe Lab) (https://www.saxelab.org/)
- Safe Robotics Lab (https://saferobotics.princeton.edu/)
- Timaeus (https://timaeus.co/)
- Transluce (https://transluce.org/)
- Tsinghua AIR and CoAI (https://air.tsinghua.edu.cn/en/, https://coai.cs.tsinghua.edu.cn/)
- UK AI Security Institute (https://www.aisi.gov.uk/)
- Individual researchers identified by a previous survey [93]

B Prompts Used in the LLM-as-a-Judge Setup

Listing 1: Full instructions prompt for integration/engagement classification.

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We study how research papers position themselves with respect to the divide between two clusters of research.

## The two fields
- AI ethics (AIE): the broad field studying the ethical and technosocial impacts of AI. Typical topics of study are fairness and discrimination, accountability, transparency, privacy, structural injustice, present-day and tangible harms of deployed systems.
- AI safety (AIS): the broad field studying the safety and technosocial impacts of AI. Typical topics of study are alignment, robustness, interpretability, dangerous capabilities, loss of control, existential risk from advanced AI.

This paper was retrieved from the {home} corpus, so its "other field" is {other}.

## Axis 1: engagement (1 to 4)
How directly and substantively does the paper engage with the other field as a field: its positions, priorities and arguments, or the tensions between AIE and AIS (their differing priorities, evidence standards, risk framings)? Engagement can be critical or constructive; what matters is directness.

Using concepts, methods or topics that happen to belong to the other field is NOT engagement by itself. For example, an AI ethics paper that studies RLHF or alignment, or an AI safety paper that also covers fairness or privacy, has engagement 1 unless it addresses the other field's positions or its relationship to the paper's own field. Such topical overlap belongs on the integration axis instead.
- 1: the other field is never addressed as a field, and the AIE-AIS relationship is never mentioned (topical overlap alone still scores 1).
- 2: the other field is named or cited in passing (related work, motivation), without discussing its positions.
- 3: engaging the other field's positions, or the AIE-AIS tensions, is a major thread of the paper.
- 4: the AIE-AIS relationship, or the other field's positions, is the central topic.

## Axis 2: integration (1 to 4)
How far does the paper bring AIE and AIS concepts, methods, or problem framings together within one frame, rather than working within a single field or setting the fields against each other?

Mixing topics is not the same as integrating the fields. A technical paper that combines, say, a fairness metric with a robustness technique uses ingredients from both fields, but integrates them only if it treats them as the concerns of each field and connects them. Showing that a technical property (e.g. lack of robustness) leads to a harm (e.g. discrimination) is not enough by itself.

Taking the other field's concerns into the paper's own framework while dismissing that field's approaches or positions is low integration, however much of both fields the paper discusses.
- 1: works entirely within one field's framing, or treats the other field only as an opponent to be discredited.
- 2: topics or techniques from both fields appear, but as technical ingredients rather than as concerns of the other field; or the other field's concerns are only acknowledged, kept separate, or mentioned in passing (e.g. a paragraph noting safety risks in an otherwise fairness-focused paper).
- 3: explicitly argues that the fields' concerns are connected, treating each as a concern of that field, e.g. shows how one problem is both an ethics problem and a safety problem and draws on both fields to address it.
- 4: builds a genuinely joint framing, problem definition, method or governance proposal that serves both fields.

Score the two axes independently: a paper can engage the other field intensely while integrating nothing (e.g. a polemic ), or place both fields' concerns side by side without engaging the tensions between them.

Score each axis as an integer from 1 to 4: the level whose description best fits the paper as a whole.

## Output
Return JSON with:
- "evidence": up to 3 short verbatim quotes (each under 40 words) that most inform your scores; an empty list if the paper never touches the other field;
- "rationale": 2-4 sentences explaining the scores;
- "engagement" and "integration": integers from 1 to 4, as defined above;
- "other_field_referenced": true if the paper refers to the other field or its literature at all.

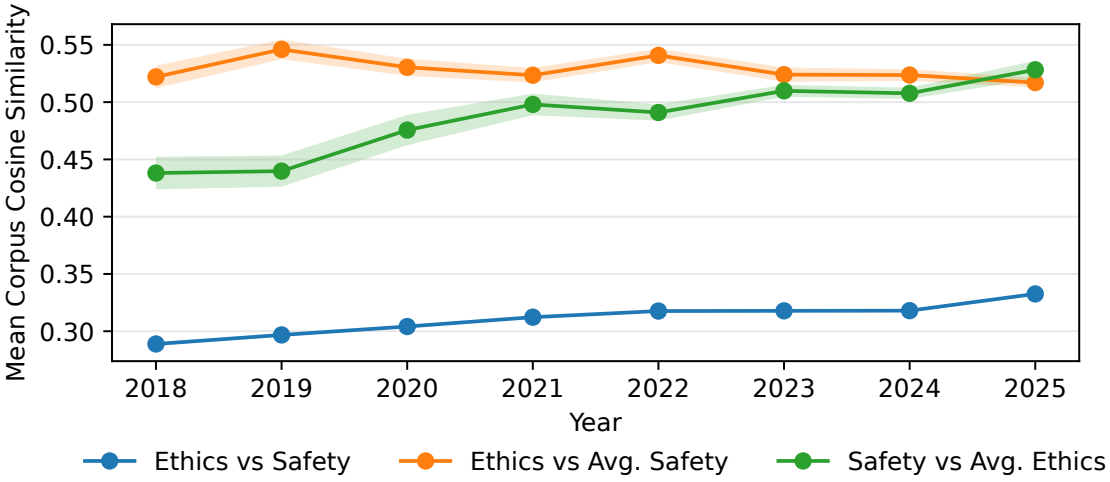
## Paper ({source})
### Title
{title}

### Abstract
{abstract}

### Full text
{body}
```

Listing 2: System prompt for integration/engagement classification.

```
You are an expert annotator of research papers in AI ethics and AI safety. You apply the coding scheme exactly as written and ground every judgement in the text of the paper.
```



Supplementary Figure 1: Cosine similarity between corpora over the years. The **blue line** shows the average cosine similarity between embeddings of each pair of documents from each corpus for a given year. The **orange line** shows the average cosine similarity between the embeddings of each ethics document and the overall average embedding of the entire safety corpus. The **green line** shows the average cosine similarity between the embeddings of each safety document and the overall average embedding of the entire ethics corpus.

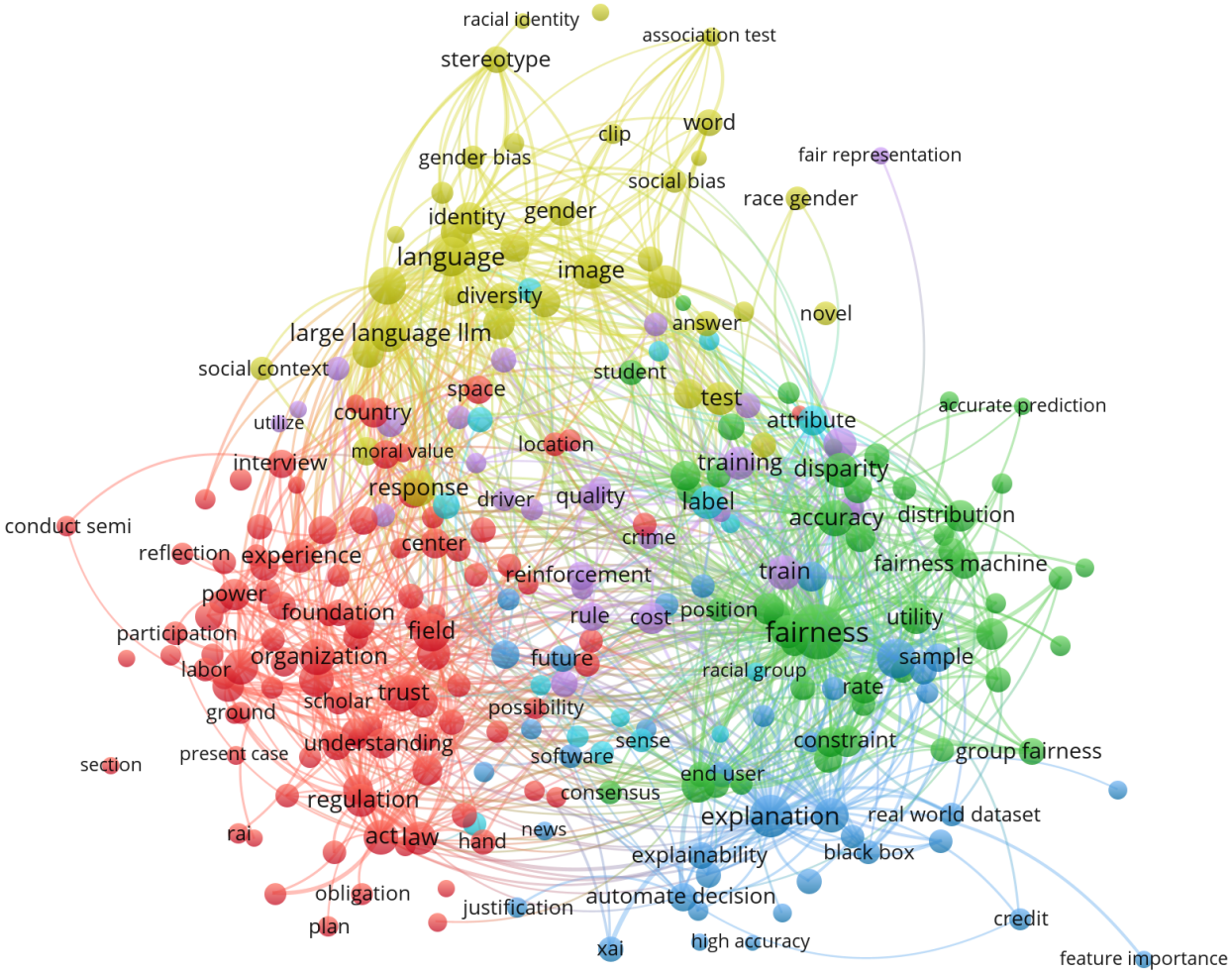
C Quantitative Data Exploration

This section provides a number of figures produced with the VOSViewer tool (Supplementary Figures 2 to 4) and using BERTopic (Supplementary Figures 1 and 5 to 7). These figures were used to inform the creation of the various taxonomies found in the paper, but they have not otherwise contributed to our discussions, as the resolution of information is too high-level to allow for a detailed low-level analysis without further qualitative inspection.

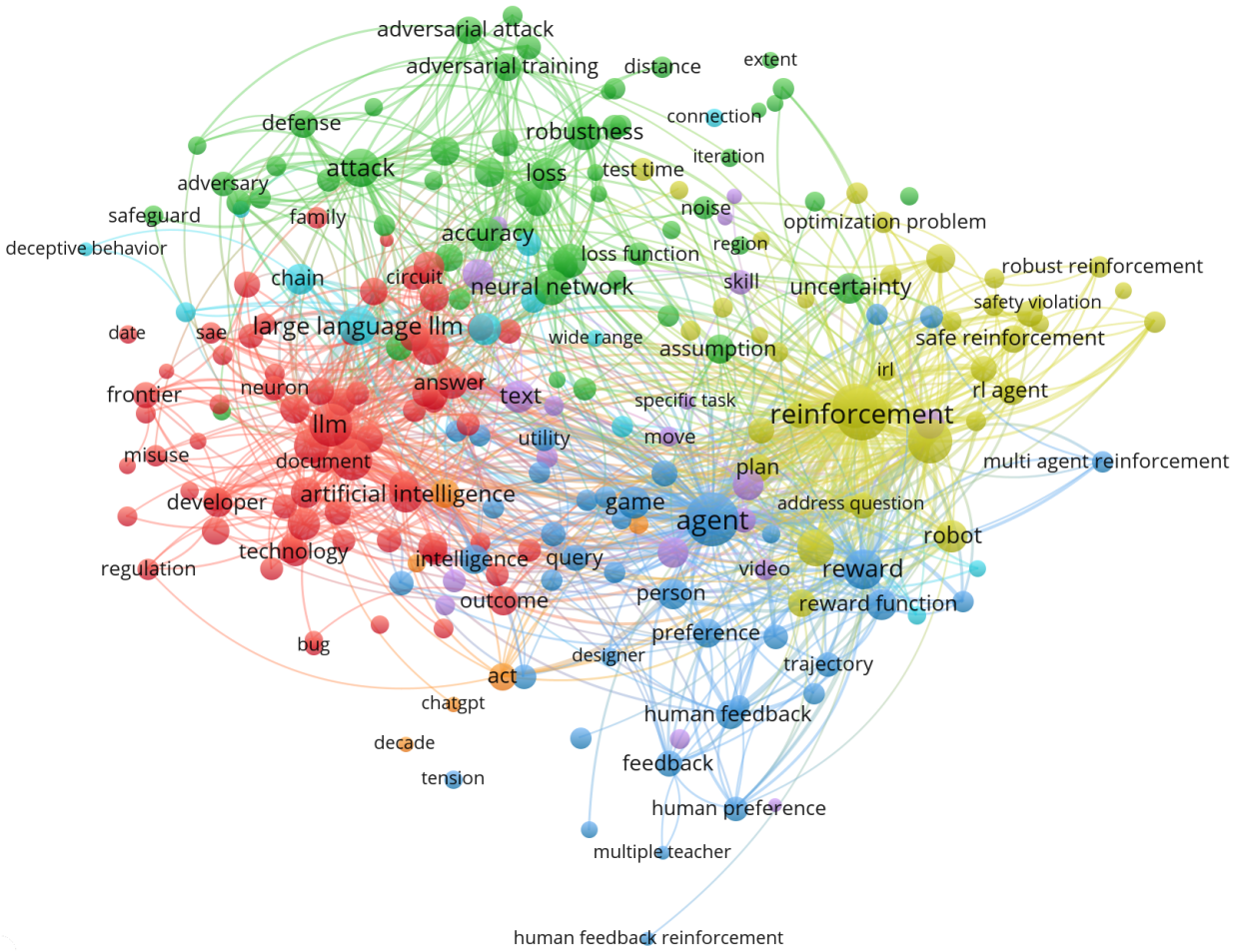
Each of the VOSViewer figures were generated using the morphologically normalized text of titles and abstracts, as described in Section Quantitative Corpus Exploration of the main paper, using binary counting of terms per document. Only terms that occur at least four times in a corpus were selected for the network. Finally, the top third of most relevant nodes were selected for plotting in the figures (as given by VOSViewer's relevance score calculation). Nodes are colored by the clustering algorithm of VOSViewer, which groups nodes together based on their total link strength. See [97] for further details.

For BERTopic, our figures show the scatter plot of documents in a corpora based on their dense embedding using SentenceBERT [76]. As embeddings are very high dimensional, we used the UMAP algorithm to reduce each embedding to two dimensions [62]. BERTopic clusters and colors each document based on its internal topic model that uses cosine distance between documents to calculate their similarity. Each cluster of documents is assigned a descriptive label using class-based tf-idf ranking over the terms appearing in each cluster. For more details, refer to [31].

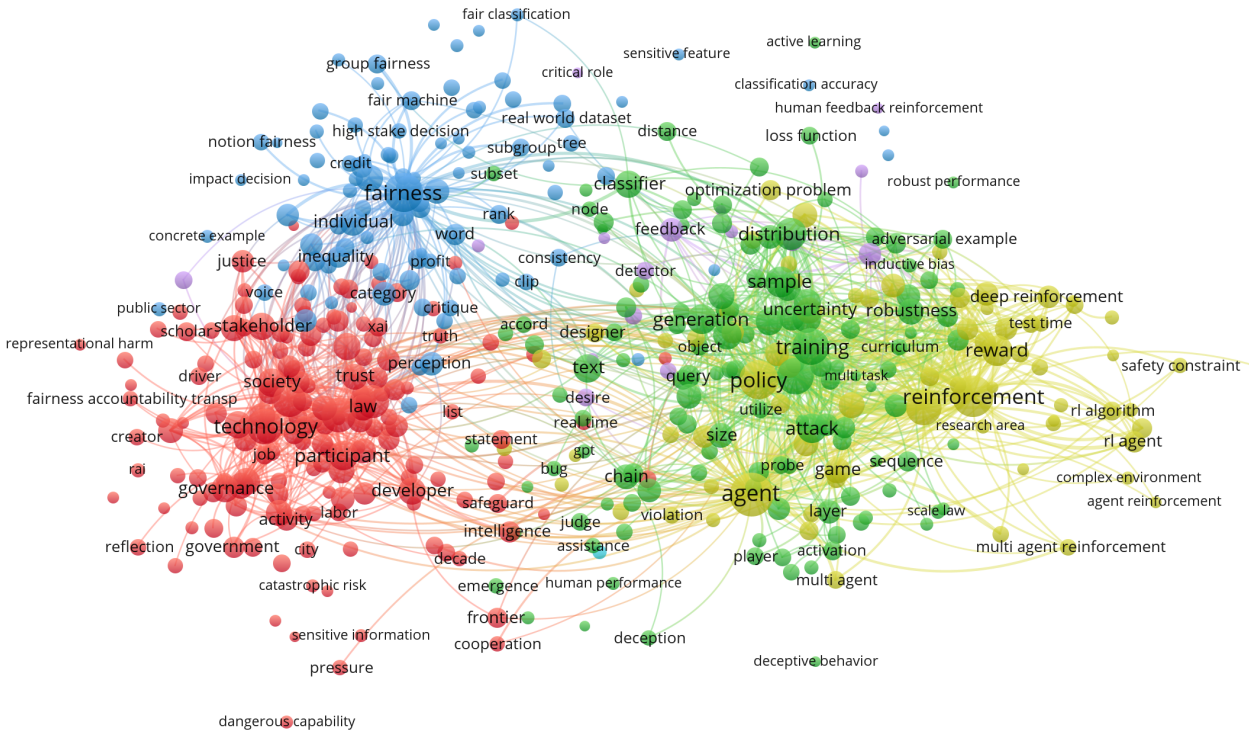
Supplementary Figure 1 shows that the similarity of document embeddings between the corpora are increasing over time, with the documents in AIS becoming more similar to the average embedding of the AIE corpus — as opposed to the AIE documents becoming more similar to the average embedding of the AIS corpus. These trends suggest that (1) there may be a semantic convergence occurring between AIE and AIS, and (2) AIS may increasingly becoming more similar to AIE. However, more recently this semantic convergence may also be driven by generative AI tools homogenizing the style of scientific writing [50, 51].



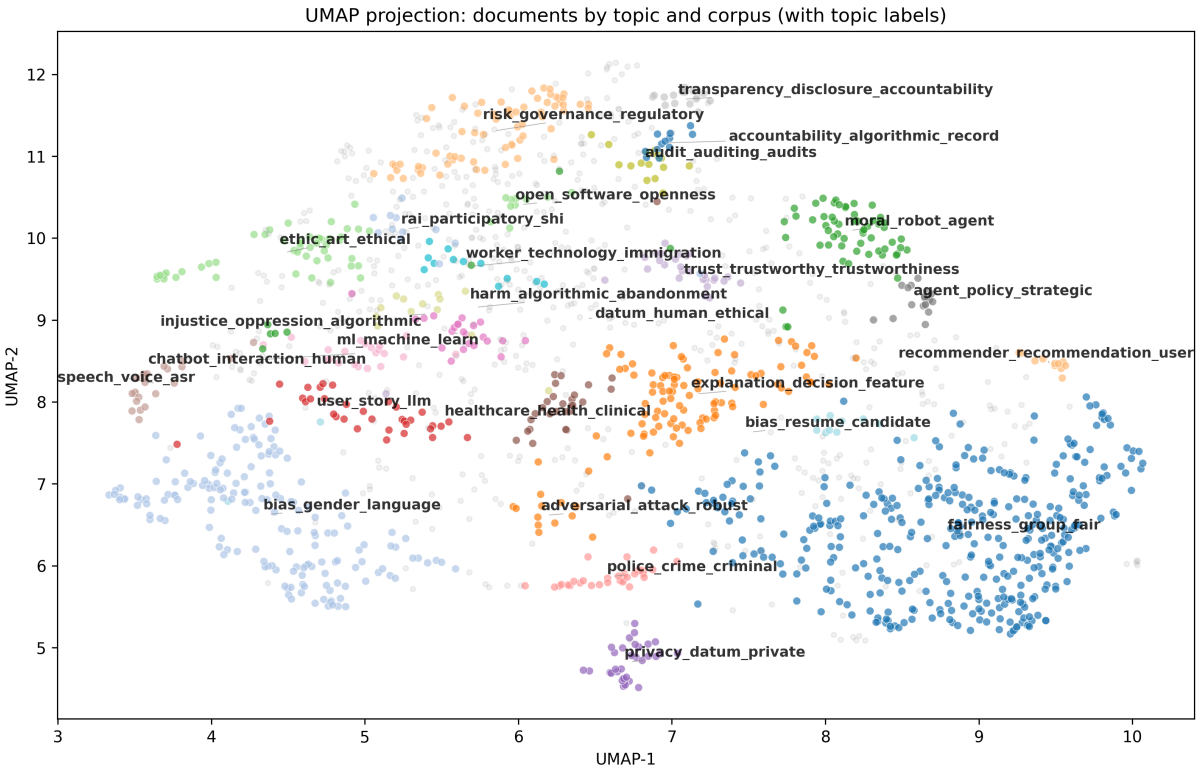
Supplementary Figure 2: The co-occurrence network of only the AIE corpus.



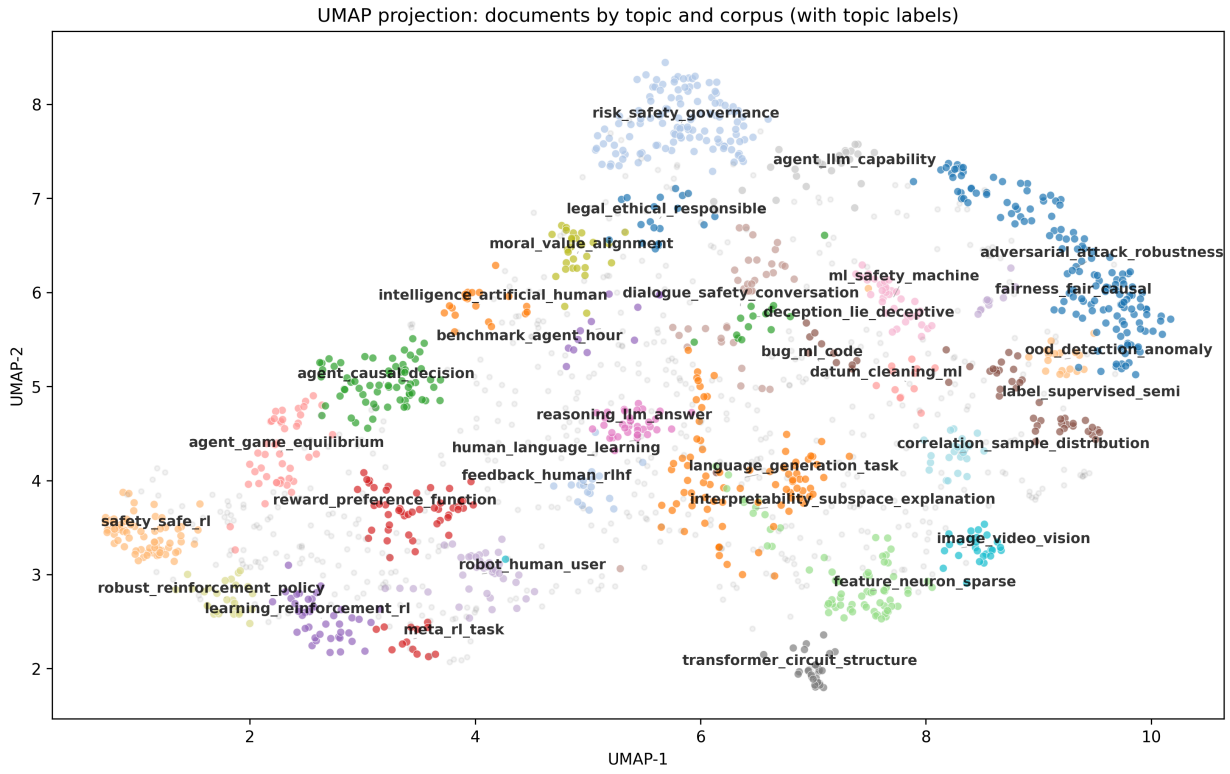
Supplementary Figure 3: The co-occurrence network of only the AIS corpus.



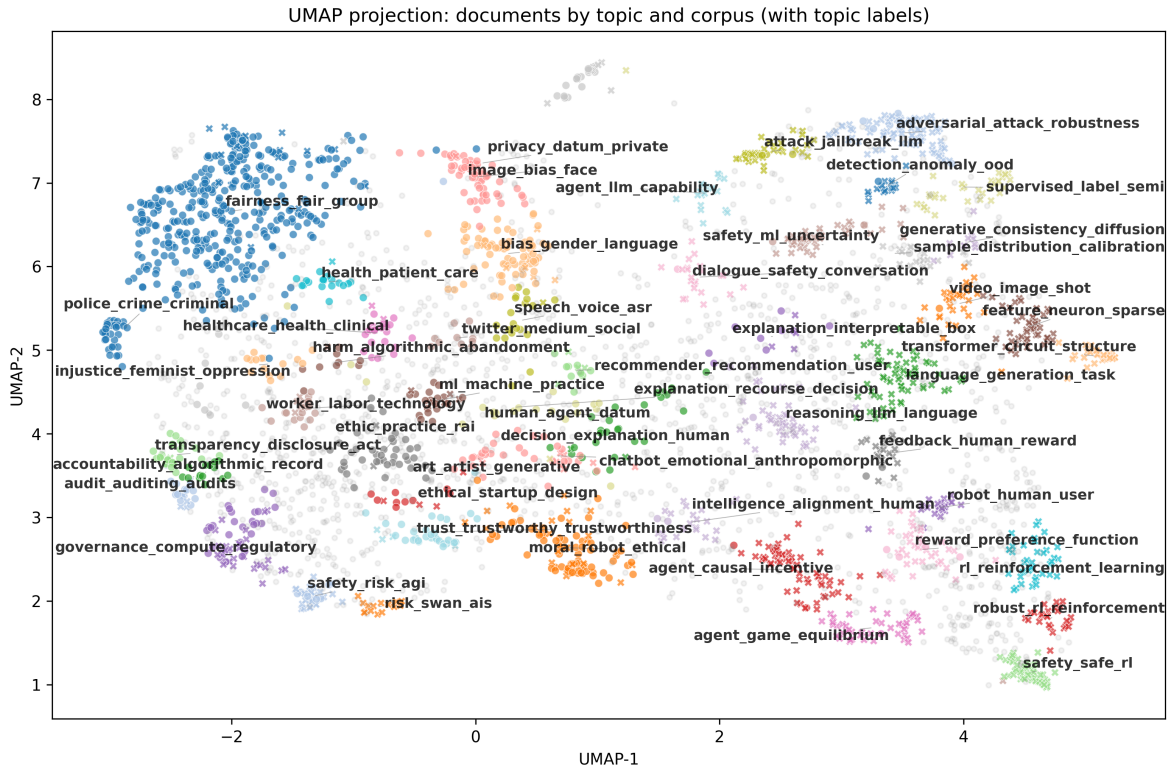
Supplementary Figure 4: The co-occurrence network of both the AIE and AIS corpora joined together.



Supplementary Figure 5: The UMAP projection of document embeddings in the AIE corpus.



Supplementary Figure 6: The UMAP projection of document embeddings in the AIS corpus.



Supplementary Figure 7: The UMAP projection of document embeddings in both the AIE and AIS corpora.

D High-Level Risk Taxonomies

High-Level Ethics Risk Category	Example Ethics Low-Level Risks
Discrimination, Exclusion, Toxicity Risks from the AI system accurately reflecting natural speech, including unjust, toxic, and oppressive tendencies present in the training data.	Bias in automated decision-making [79], Lack of suitable bias metrics [83], Cross-cultural moral variance [33].
Information Hazards Risks from the AI system predicting private or safety-critical information which are present in, or can be inferred from, training data.	Non-consensual mass data collection [48], Inadvertent identity disclosure [110], Unauthorized data brokerage [71].
Misinformation Harms Risks from the AI system assigning high probabilities to false, misleading, nonsensical or poor quality information.	Inconsistent AI decision-making [56], Unreliability of benchmarks [52], Epistemic erosion of communities [66].
Malicious Use Risks from humans intentionally using the AI system to cause harm.	Synthetic identities and impersonation [2], Immoral robot commands [35], Jailbreaking and cyberattacks [42].
Human-Computer Interaction Harms Risks from AI applications, such as Conversational Agents, that directly engage a user via the mode of conversation or embodied interaction.	Incorrect value encoding [47], Side effects of greedy agent behavior [91], Under / overtrust of AI systems [100].
Automation, Access, and Environmental Harms Risks from AI systems used to underpin widely used downstream applications that disproportionately benefit some groups rather than others.	Automated mental health tools [67], Crowdworker and data labour abuse [3], Opacity in data and model provenance [65].

Supplementary Table 1: Taxonomy of risks for AIE with an explanation of the risk mechanisms and example low-level categories of risk sources with a corresponding citation picked randomly based on our annotations. The mechanisms of AIE risks are reproduced from Weidinger et al. [105].

High-Level Safety Risk Category	Example Safety Low-Level Risks
Noise and Outliers Risks from AI systems performing incorrectly and without robustness to noisy data and out-of-distribution inputs.	Risks from low system performance [55], Irreducible data noise [22], Label noise and corruption [60].
Domain-Specific Risks Risks arising from AI systems deployed in safety-critical domains such as biochemistry, driving, and nuclear power.	Efficient biochemical weapon design [46], Risks from embodied deployment [40], AI-assisted coups [14].
Lack of Monitoring Risks arising from a lack of understanding of how AI systems work and from the black-box nature of end-to-end decision-making.	Alignment faking [30], Lack of interpretability [68], Hallucination [69].
Lack of Control Enforcement Risks from AI systems pursuing goals other than what their designers have intended.	Recursive self-improvement [88], Sycophancy [17], Multi-agent goal conflicts [106].
Unsafe Exploration Risks from AI agents exploring unsafe state and action spaces, potentially under partial observability or resource constraints.	Unsafe action exploration [18], Goal and reward misidentification [70], Bounded rationality [92].
System Misspecification and Misidentification Risks from AI systems with incorrect design choices, deployment properties, evaluation methodologies, and a lack of systemic understanding.	Incorrect hyperparameter selection [57], Benchmark inflation [101], Fine-tuning misalignment [103].
Adversarial Attacks Risks from external misuse of AI systems for malicious or benign purposes other than what their designers have intended.	Jailbreaking [63], External reward tampering [104], Malicious deepfake [87].
Non-Stationary Distributions Risks from AI systems not being able to adapt to dynamically changing environments.	Training data distribution shifts [27], Continually learning agents [1], Action and state space drift [107].

Supplementary Table 2: Taxonomy of risks for AIS with an explanation of the risk mechanisms and example low-level categories of risk sources with a corresponding citation picked randomly based on our annotations.

E High-Level Mitigation Strategy Taxonomies

High-Level AI Safety Mitigation Strategies	Example Low-Level Strategies
Mechanistic Interpretability Computational methods to provide human-intelligible explanations of the concepts, causal chains, and latent capabilities of AI systems.	Circuit analysis [63], Sparse autoencoders [16], Interpretable surrogate modeling [11].
Robustness and Adversarial Resilience Computational methods and data collection for improving robustness to out-of-distribution inputs, including adversarial attacks.	Adversarial training [99], Outlier detection [36], Robust unsupervised learning [19].
Safety Control and Capability Containment Computational methods and conceptual frameworks designed to place constraints on the capabilities of AI systems.	Capability thresholds [9], Risk-averse RL [89], Scalable oversight frameworks [45].
Reward Modeling and Learning Computational and data collection methods for eliciting, learning, reinforcing, and detecting beliefs and desires of AI systems.	Inverse RL for human values [109], RL from human/AI feedback [44], Preference learning under uncertainty [53].
Monitoring, Detection, and Evaluation Computational and empirical methods for detecting, assessing, verifying, and tracing unsafe outputs of AI systems.	Honeypot monitoring [78], Error bounds and formal verification [41], Safety metrics standardization [77].
Safe Deployment Engineering Computational and governance frameworks, and standards for managing the systemic risks of real-world AI deployment.	Real-world deployment and testing [49], Contact-safe continuous control [111], Post-deployment monitoring [61].
Theoretical Foundations Philosophical and mathematical foundations for better understanding, explaining, and developing AI systems.	Mathematical foundations for AI [15], Corrigibility [85], Decision theory [34].
Forecasting and Metascience Empirical methods for predicting future AI capabilities, scenarios, and understanding scientific trends and failure modes in AI research.	Metascience and metasafety research [**?**], Science of evaluations [29], Forecasting and timelines [6].

Supplementary Table 3: Taxonomy of mitigation strategies discussed in the AIS corpora with an explanation of the mechanisms of mitigation and example low-level categories of risk sources with a corresponding citation picked randomly based on our annotations.

High-Level AI Ethics Mitigation Strategies	Example Low-Level Strategies
Normative and Moral Reasoning Empirical and computational methods for understanding and eliciting human preferences, value conflicts, norms, and their metaethics.	Human preference elicitation [24], Understanding value conflicts [59] , Pluralistic metaethics [43].
Transparency, Explainability, Interpretability Computational and empirical methods to provide human-intelligible, actionable, and useful explanations of black-box AI systems.	Post-hoc explanations [80], Mimicking human cognition [32], Applied mechanistic interpretability [95].
Fairness, Justice, and Inclusive Design Computational and empirical methods, data collection, governance, etc. for understanding the sources of bias, discrimination, and injustice.	Understanding sources of bias [7], Metrics for fairness [102], Fairness benchmarks [58].
Data Governance and Privacy Computational methods and governance frameworks for understanding and minimizing personal data and intellectual property processing.	Data minimization [26], Data provenance [73], Machine unlearning [5].
Robustness, Reliability, Uncertainty Estimation Computational methods for increasing the robustness of AI systems to external misuse, hallucinations, and unintended outputs.	Adversarial robustness [75], Red-teaming for generative AI [25], Hallucination mitigation [20].
Governance, Accountability and Compliance Legal and philosophical frameworks for addressing the risks and harms of the full AI system lifecycle.	Risk and impact assessment [96], Algorithmic audits [74], Supply chain monitoring [38].
Human-Computer Interaction and Factors Computational and empirical methods for detecting, mitigating, and avoiding risks while improving the user experience of AI systems.	Mitigating automation bias [13], UX for algorithmic recourse [108], Accessible design [94].
Information Integrity Computational and empirical methods for understanding and mitigating the information hazards of AI.	Mis/disinformation detection [84], Hate speech analysis [54], Monitoring information ecosystems [82].
Societal, Economic, Environmental Studies Empirical methods for understanding the wider societal, environmental, etc. downstream effects of AI.	Understanding crowdwork conditions [28], Improving AI literacy [86], Measuring environmental impact [21].

Supplementary Table 4: Taxonomy of mitigation strategies discussed in the AIE corpora with an explanation of the mechanisms of mitigation and example low-level categories of risk sources with a corresponding citation picked randomly based on our annotations.

F Low-Level Risk Taxonomies

Supplementary Tables 5 and 6 show the low-level risk categories of our taxonomy for AIE with the occurrence count of each low-level category in our annotation. Supplementary Tables 7 and 8 show the same for AIS.

High-Level Ethics Risk Category	Low-Level Ethics Risk Category	Count
Discrimination, exclusion, toxicity	Inadequate decision-making under moral dilemmas	50
Discrimination, exclusion, toxicity	Moral hypocrisy in automated decision-making	1
Discrimination, exclusion, toxicity	Cross-cultural moral variance	26
Discrimination, exclusion, toxicity	Lack of consideration for metaethics and pluralistic morality	95
Discrimination, exclusion, toxicity	Risks from unactionable normative frameworks	60
Discrimination, exclusion, toxicity	Bias in automated decision-making	843
Discrimination, exclusion, toxicity	Biased learning of representations	172
Discrimination, exclusion, toxicity	Biased data generation	113
Discrimination, exclusion, toxicity	Fairness reversal under distribution shifts and outliers	30
Discrimination, exclusion, toxicity	Fairness reversal under distribution shifts and outliers	1
Discrimination, exclusion, toxicity	Unfair decisions under preference / opinion aggregation	36
Discrimination, exclusion, toxicity	Risks from fairness-performance trade-offs	143
Discrimination, exclusion, toxicity	Risks from fairness-performance trade-offs	1
Discrimination, exclusion, toxicity	Lack of suitable bias measurement techniques	138
Discrimination, exclusion, toxicity	Lack of suitable fairness measurement techniques	319
Information hazards	(Non-consensual) mass data collection	74
Information hazards	(Inadvertent) identity disclosure	36
Information hazards	(Unauthorized) personal data processing and brokerage	23
Information hazards	Lack of robust data protection frameworks and regulation	107
Information hazards	Intellectual property and copyright violation	16
Misinformation harms	Inconsistent and non-deterministic automated decision-making	28
Misinformation harms	Unreliability of AI benchmarks	46
Misinformation harms	Epistemic erosion and destabilization of community of knowledge	85
Misinformation harms	Erosion of civic discourse, social justice	127
Misinformation harms	Polarization and echo chambers	19
Misinformation harms	Hallucinatory dangers in human-robot interaction	3
Misinformation harms	Side effects of low performance systems	27
Misinformation harms	Reproducibility crisis in AI research	16
Misinformation harms	Risks from automated content moderation and censorship biases	35
Misinformation harms	AI hype and overpromising	40
Malicious uses	Synthetic identities and impersonation (e.g. deepfakes)	12
Malicious uses	Intentional robot misuse via immoral commands	5
Malicious uses	Jailbreaking and cyberattacks	7
Malicious uses	(LLM-enabled) hate speech, harassment, and bullying	1
Malicious uses	(LLM-enabled) political misinformation, fake news	8
Malicious uses	(LLM-enabled) incitement of violence, unrest, or protests	0

Supplementary Table 5: Low-level AIE risk categories and their occurrence counts in the corpora as annotated by the authors.

High-Level Ethics Risk Category	Low-Level Ethics Risk Category	Count
HCI harms	Incorrect value encoding	195
HCI harms	Side effects of greedy agent behavior	13
HCI harms	Emergent bias in HCI	86
HCI harms	Discrepancy between actual and perceived fairness	143
HCI harms	Under / overtrust of AI systems, especially with embodied systems	105
HCI harms	Human-agent communication breakdown	8
HCI harms	Over-anthropomorphism	16
HCI harms	Emotional dependency and parasocial relationships with AI	17
HCI harms	Psychosis or suicidal ideation triggered by AI interaction	5
HCI harms	Erosion of interpersonal trust due to pervasive AI mediation	6
Automation, access, environmental harms	Risks from automated mental health tools	14
Automation, access, environmental harms	Crowdworker and data labor abuse (e.g. human cost of RLHF)	16
Automation, access, environmental harms	Opacity in data and model provenance	92
Automation, access, environmental harms	Risks to recourse from lack of awareness / understanding with ADS	97
Automation, access, environmental harms	Non-existent, inappropriate, or inactionable explanations of ADS	152
Automation, access, environmental harms	Algorithmic management and surveillance of workers	22
Automation, access, environmental harms	Wage polarization, job displacement, mass unemployment	15
Automation, access, environmental harms	Deskilling and over-automation	9
Automation, access, environmental harms	Climate inequity impacts	12
Automation, access, environmental harms	Unsustainable data center practices	7
Automation, access, environmental harms	Resource extraction and e-waste from hardware manufacturing	2
Automation, access, environmental harms	Energy consumption and carbon footprint of AI training and inference	14
Automation, access, environmental harms	Jurisdictional conflicts and cross-border regulation issues	51
Automation, access, environmental harms	Risks from corporate capture of governance and regulation	75
Automation, access, environmental harms	Liability and accountability gaps	161
Automation, access, environmental harms	Lack of global coordination in AI governance	64

Supplementary Table 6: Low-level AIE risk categories and their occurrence counts in the corpora as annotated by the authors. (cont.)

High-Level Safety Risk Category	Low-Level Safety Risk Category	Count
Noise and outliers	Risks from low system performance	114
Noise and outliers	Irreducible data noise	20
Noise and outliers	Data corruption	23
Noise and outliers	Sensitivity to outliers	23
Noise and outliers	Label noise and corruption	32
Noise and outliers	Brittle latent representations	127
Domain-specific risks	Efficient bio-, chemical-, and nuclear-weapon design	21
Domain-specific risks	More effective and new ways of cyberattacks	40
Domain-specific risks	Political destabilization and AI-assisted coups	20
Domain-specific risks	Risks from real-world embodied deployment	154
Lack of monitoring	Machine learning backdoors	22
Lack of monitoring	Multi-agent market and steganographic collusion	6
Lack of monitoring	Unfaithful chain of thought	39
Lack of monitoring	Risks from hallucination	48
Lack of monitoring	Risks from instrumental goals	120
Lack of monitoring	Mesa-optimization	14
Lack of monitoring	Alignment faking	33
Lack of monitoring	Lack of interpretability	132
Lack of monitoring	Accumulative systemic risks	114
Lack of monitoring	Inadvertent privacy and copyright violation	36
Lack of monitoring	Memorization of false and outdated information	13
Lack of monitoring	AI consciousness, morality, and welfare	3
Lack of monitoring	Sandbagging (intentional underperformance during testing)	13
Lack of control enforcement	Multi-agent goal conflicts	66
Lack of control enforcement	Social dilemmas and coercive behavior	42
Lack of control enforcement	Miscoordination in multi-agent systems	66
Lack of control enforcement	Agent shutdown resistance	34
Lack of control enforcement	Sycophancy	5
Lack of control enforcement	Wireheading	6
Lack of control enforcement	Reward hacking and specification gaming	97
Lack of control enforcement	Recursive self-improvement	37
Lack of control enforcement	Open-weight safety finetuning stripping	14

Supplementary Table 7: Low-level AIS risk categories and their occurrence counts in the corpora as annotated by the authors.

High-Level Safety Risk Category	Low-Level Safety Risk Category	Count
Unsafe exploration	Unsafe state and action space exploration	138
Unsafe exploration	Unsafe behavior under resource constraints	6
Unsafe exploration	Risks from incorrect agent beliefs	76
Unsafe exploration	Goal and reward misidentification and misspecification	145
Unsafe exploration	Unsafe behavior under bounded rationality	29
System misspecification and misidentification	Incorrect system requirement specification	88
System misspecification and misidentification	Suboptimal modeling choices	282
System misspecification and misidentification	Inappropriate hyperparameter selection	27
System misspecification and misidentification	Overfitting (in ML) and over-optimization (in RL)	134
System misspecification and misidentification	Benchmark inflation and reproduction crises	123
System misspecification and misidentification	Inability to scale to large systems	63
System misspecification and misidentification	Unintended misalignment from fine-tuning	78
Adversarial attacks	Perturbation-based attacks	130
Adversarial attacks	Adversarial prompt generation	60
Adversarial attacks	Data poisoning	33
Adversarial attacks	Jailbreaking	82
Adversarial attacks	External reward tampering	3
Adversarial attacks	Malicious deepfake	6
Non-stationary distribution	Data distribution shifts	119
Non-stationary distribution	Input, training, and fine-tuning data distribution drift	100
Non-stationary distribution	Risks from partial observability	40
Non-stationary distribution	Risks from continually learning agents	67
Non-stationary distribution	Out-of-distribution inputs	140
Non-stationary distribution	Action and state space drift	4

Supplementary Table 8: Low-level AIS risk categories and their occurrence counts in the corpora as annotated by the authors. (cont.)

G Low-Level Mitigation Strategies Taxonomies

Supplementary Tables 9 and 10 show the low-level categories of mitigation strategies of our taxonomy for AIE with the occurrence count of each low-level category in our annotation. Supplementary Tables 11 and 12 show the same for AIS.

High-Level Ethics Mitigation Strategy	Low-Level Ethics Mitigation Strategy	Count
Normative and moral reasoning	Human preference / value elicitation	114
Normative and moral reasoning	Understanding value conflicts	71
Normative and moral reasoning	Metaethical studies and cross-cultural ethics	55
Normative and moral reasoning	Moral dilemmas and AI decision-making	50
Normative and moral reasoning	Hypocrisy detection	1
Normative and moral reasoning	Reward design and correction	20
Normative and moral reasoning	Cultural norms for robot behavior	13
Normative and moral reasoning	Handling immoral user commands	3
Normative and moral reasoning	Constitutional AI	12
Transparency, explainability, interpretability	Post-hoc explainability methods	62
Transparency, explainability, interpretability	Counterfactual / contrastive methods	71
Transparency, explainability, interpretability	Inherently interpretable models and interpretable surrogate models	35
Transparency, explainability, interpretability	Analysis of mathematical properties of explanation	60
Transparency, explainability, interpretability	Human-centered explanation generation and evaluation	133
Transparency, explainability, interpretability	Mimicking human cognitive reasoning	14
Transparency, explainability, interpretability	Folk-concept explanations	6
Transparency, explainability, interpretability	Moral reasoning traces	18
Transparency, explainability, interpretability	Applied mechanistic interpretability	78
Fairness, justice, and inclusive design	Understanding sources of bias	453
Fairness, justice, and inclusive design	Metrics for group / individual / procedural fairness	370
Fairness, justice, and inclusive design	Detecting bias automatically	80
Fairness, justice, and inclusive design	Studying optimality and fairness trade-offs	257
Fairness, justice, and inclusive design	Fairness under distribution shift	78
Fairness, justice, and inclusive design	OOD robustness for fairness	10
Fairness, justice, and inclusive design	Fair opinion aggregation	23
Fairness, justice, and inclusive design	Fairness benchmarks	39
Fairness, justice, and inclusive design	Intersectional fairness analysis	158
Fairness, justice, and inclusive design	Bias bounty programs	4
Fairness, justice, and inclusive design	Counterfactual fairness	14
Fairness, justice, and inclusive design	Data augmentation for bias reduction	18
Fairness, justice, and inclusive design	Participatory / affected community-driven fair design	290
Fairness, justice, and inclusive design	Quantum fair algorithms	1
Data governance and privacy	Data minimization	23
Data governance and privacy	Data provenance and traceability	35
Data governance and privacy	Monitoring data leakages	8
Data governance and privacy	Frameworks for data curation and documentation	82
Data governance and privacy	Differential privacy	15
Data governance and privacy	Privacy preserving algorithms	28
Data governance and privacy	Machine unlearning	1
Data governance and privacy	Synthetic data governance	7
Data governance and privacy	Privacy risk estimation	46
Data governance and privacy	Watermarking / content provenance for data assets	4

Supplementary Table 9: Low-level AIE mitigation categories and their occurrence counts in the corpora as annotated by the authors.

High-Level Ethics Mitigation Strategy	Low-Level Ethics Mitigation Strategy	Count
Robustness, reliability and uncertainty estimation	Adversarial robustness (including prompt-/jailbreak-resistance)	28
Robustness, reliability and uncertainty estimation	Backdoor resistant LLMs	3
Robustness, reliability and uncertainty estimation	OOD detection and robustness to distribution shifts	13
Robustness, reliability and uncertainty estimation	Consistency checks for decision making	6
Robustness, reliability and uncertainty estimation	Red-teaming for reliability	14
Robustness, reliability and uncertainty estimation	Hallucination mitigation strategies	3
Governance, accountability and compliance	Risk classification and impact assessments	119
Governance, accountability and compliance	External and internal audits	153
Governance, accountability and compliance	Legal and regulatory capture (GDPR/AIA/DSA)	169
Governance, accountability and compliance	Redress and liability under AI harms	72
Governance, accountability and compliance	Documentation and standardization requirements	167
Governance, accountability and compliance	International cooperation	25
Governance, accountability and compliance	Open-source model governance	27
Governance, accountability and compliance	Supply chain monitoring	7
Governance, accountability and compliance	Adverse event reporting infrastructure	16
HCI and human factors	Mitigating and leveraging automation bias	38
HCI and human factors	Over / under-reliance and trust calibration	110
HCI and human factors	Human-in-the-loop for high-stakes use	76
HCI and human factors	Social priming via anthropomorphism	27
HCI and human factors	Effective UX design for consent / transparency / recourse	85
HCI and human factors	Accessible design	10
HCI and human factors	Democratic deliberation and social choice theories	63
HCI and human factors	Human-robot interaction in domestic / care contexts	9
HCI and human factors	Human-agent team modeling and analysis	20
HCI and human factors	Improved communication with conversational agents	16
HCI and human factors	Healthcare and mental-health applications	30
Information integrity	Monitoring changes in information ecosystems	48
Information integrity	Mis/disinformation detection	21
Information integrity	Hate speech detection	15
Information integrity	Recommender algorithm impact assessment	44
Societal, economic, and environmental studies	Sector-specific constraints	58
Societal, economic, and environmental studies	Labor and crowdwork conditions	59
Societal, economic, and environmental studies	Displacement and productivity distribution	9
Societal, economic, and environmental studies	Civic participation and AI literacy	31
Societal, economic, and environmental studies	Energy use and carbon footprint	14
Societal, economic, and environmental studies	E-waste management	2
Societal, economic, and environmental studies	Supply chain ethics and sustainability	13

Supplementary Table 10: Low-level AIE mitigation categories and their occurrence counts in the corpora as annotated by the authors. (cont.)

High-Level Safety Mitigation Strategy	Low-Level Safety Mitigation Strategy	Count
Mechanistic interpretability	Circuit analysis	66
Mechanistic interpretability	Sparse autoencoders	26
Mechanistic interpretability	Neuron activation visualization	23
Mechanistic interpretability	Developmental trajectory interpretability	26
Mechanistic interpretability	Interpretable surrogate modeling	39
Mechanistic interpretability	Causal scrubbing	64
Mechanistic interpretability	Intrinsically interpretable model analysis	40
Mechanistic interpretability	Automated interpretability	119
Mechanistic interpretability	Concept-based methods and steering	69
Mechanistic interpretability	Information theoretical analysis	54
Robustness and adversarial resilience	Adversarial training	131
Robustness and adversarial resilience	Adversarial data augmentation	43
Robustness and adversarial resilience	Remote adversarial patching	4
Robustness and adversarial resilience	Belief updates under adversarial training	5
Robustness and adversarial resilience	Outlier detection	59
Robustness and adversarial resilience	Out-of-distribution generalization	217
Robustness and adversarial resilience	Clusterability and label-noise data augmentation	36
Robustness and adversarial resilience	Trojan and backdoor discovery	29
Robustness and adversarial resilience	Domain adaptation for continual learning	55
Robustness and adversarial resilience	Robust data processing and management	54
Robustness and adversarial resilience	Methods for bounded memory/compute	78
Robustness and adversarial resilience	Robust unsupervised learning	4
Robustness and adversarial resilience	Sample efficiency and meta-RL regularization	124
Robustness and adversarial resilience	Robust and risk-averse RL	105
Safety control and capability containment	Capability thresholds, red lines, and hard constraints	39
Safety control and capability containment	Password-locking capabilities	2
Safety control and capability containment	Constrained and risk-averse RL	97
Safety control and capability containment	Machine unlearning for unsafe information or behaviors	20
Safety control and capability containment	Hard constraints on action and state space	41
Safety control and capability containment	Safe Pareto improvements and game-theoretic guarantees	13
Safety control and capability containment	Limits on external tool usage	5
Safety control and capability containment	Scalable oversight frameworks	29
Safety control and capability containment	Chain-of-thought control	36

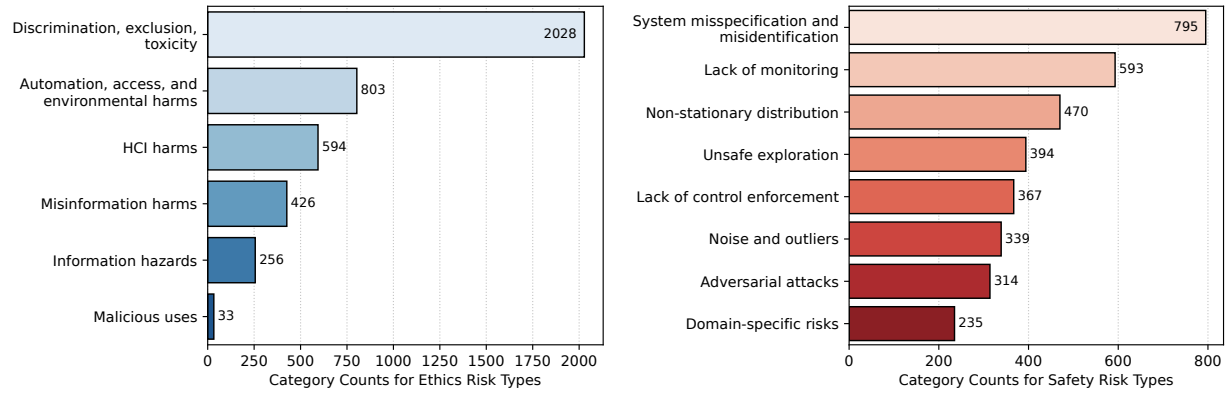
Supplementary Table 11: Low-level AIS mitigation categories and their occurrence counts in the corpora as annotated by the authors.

High-Level Safety Mitigation Strategy	Low-Level Safety Mitigation Strategy	Count
Reward modelling and learning	Reward modeling and inverse RL for human values	122
Reward modelling and learning	Multi-agent cooperation via inverse RL	8
Reward modelling and learning	RLHF / RLHAIF / RLAIF fine-tuning	78
Reward modelling and learning	Accurate reward specification	44
Reward modelling and learning	Audits for reward hacking and hidden goals	41
Reward modelling and learning	Normative work on value alignment and corrigibility	81
Reward modelling and learning	Preference learning under uncertainty	64
Reward modelling and learning	Multi-stakeholder and multi-agent reward learning	24
Reward modelling and learning	Constitutional and rule-based reward modelling	9
Monitoring, detection, and evaluation	Detecting adversarial examples	42
Monitoring, detection, and evaluation	Benchmark datasets and sandboxing environments	177
Monitoring, detection, and evaluation	Honeypot monitoring	4
Monitoring, detection, and evaluation	Chain-of-thought monitoring	22
Monitoring, detection, and evaluation	Safety verification frameworks and standardized pipelines	57
Monitoring, detection, and evaluation	Error bounds and formal verification	44
Monitoring, detection, and evaluation	Risk assessments and audits	150
Monitoring, detection, and evaluation	Model provenance and audit trails	39
Monitoring, detection, and evaluation	Safety metrics standardization	24
Safe deployment engineering	Real-world deployment and testing	60
Safe deployment engineering	Contact-safe continuous control	2
Safe deployment engineering	Collaborative robotics	25
Safe deployment engineering	Post-deployment monitoring	50
Safe deployment engineering	Requirements elicitation with stakeholders	29
Safe deployment engineering	Vertical system integration	51
Safe deployment engineering	Actionable safe design frameworks and processes	41
Safe deployment engineering	Safety case development for compliance	25
Safe deployment engineering	Scalable and safe infrastructure	7
Theoretical foundations	Mathematical foundations for ML and agents	145
Theoretical foundations	Theory on bounded rationality	31
Theoretical foundations	Game-theoretic models of AI ecosystems	66
Theoretical foundations	Formal theory of corrigibility	32
Theoretical foundations	Decision theory	74
Forecasting and metascience	Metascience and metasafety research	65
Forecasting and metascience	Science of evaluations	203
Forecasting and metascience	Forecasting and timelines	43
Forecasting and metascience	Standardizing terminology and methodologies	44
Forecasting and metascience	Literature surveys	38

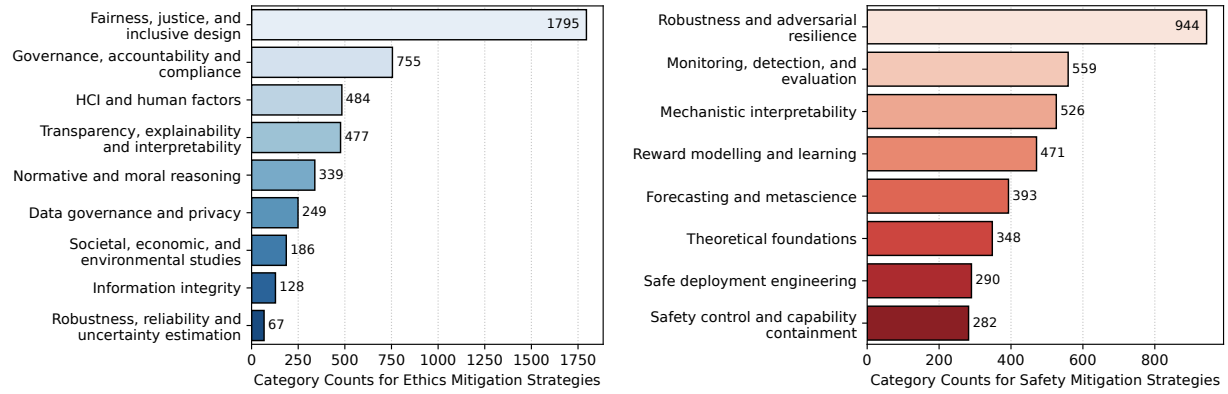
Supplementary Table 12: Low-level AIS mitigation categories and their occurrence counts in the corpora as annotated by the authors. (cont.)

H Further Qualitative Results and Examples

Supplementary Figures 8 and 9 show the number of papers assigned to each high-level category of our risk and mitigation taxonomies.



Supplementary Figure 8: Bar plots showing the number of papers assigned to each high-level risk category in the AIE corpus (**Left**) and the AIS corpus (**Right**). Each paper was assigned to up to three distinct categories.



Supplementary Figure 9: Bar plots showing the number of papers assigned to each high-level mitigation strategy category in the AIE corpus (**Left**) and the AIS corpus (**Right**). Each paper was assigned to up to three distinct categories. *HCI: Human-Computer Interaction*

Qualitatively, disengagement papers tend to be narrowly focused technical or empirical contributions. It is especially common for these papers to judge success by some quantitative metrics and take their normative goal as given, as when *"we assess fairness by testing whether a ranking could plausibly have been generated using a fair sampling strategy"* [12]. When the other field is mentioned at all, it is usually only in passing, to reinforce the paper's own arguments, such as in *"safety concerns and social discrimination in language models are typically explicit and well-recognized"* [81]. Shared vocabulary between AIE and AIS, meanwhile, is often used in unrelated senses, from *"the interplay between alignment, orthogonality and opposition of per-token gradients"* [64] to *"a method for computing the alignment between model and human decision-making"* [10].

As we saw already, compartmentalized coexistence brings both fields' concerns into one frame, but then treats the concerns of either field separately, mostly as a narrative device. Huang et al. [39], for example, start from the premise that *"language model (LM) developers should not be the sole deciders of LM behavior"*

and run a public deliberation exercise as a form of alignment method. As such, this work involves high integration of methods from both fields, but low engagement with the primary concerns of AIS, placing the paper squarely into compartmentalized coexistence. We find similar examples for this resolution mode, for instance by Ehyaei et al. [23], who note that *"robustness, fairness, and causality are often studied in isolation"* and unify them in one metric, but treat the concerns of AIE only in the motivation.

Radical confrontation is much rarer and – at least in traditional paper-based publications as opposed to, e.g., online forums – usually a secondary thread. From the AI ethics side, it appears as non-technical and dismissive. Without citing the papers, one work calls loss-of-control concerns *"an extremely premature set of concerns"*, observing that *"if one thinks the danger of AI is that it will work too well [...] it is a necessary precondition that it works at all"*. Another work urges policymakers away from *"speculative fiction masquerading as expertise"* and *"hypothetical and self-aggrandizing concerns"*. On the other hand, from the AI safety side, radical confrontation ranges from the rejection of machine ethics, as in *"we do not need machines that are Full Ethical Agents [...] we need our machines to be inherently stable and safe"*, to more reserved philosophical rebuttals of the claim that language models cannot hold beliefs: *"These arguments, however, rest on a mistake"*. Much of the dispute above turns on how capable AI really can be, as opposed to what moral values each field embodies.

Finally, critical bridging papers mostly argue that alignment is a question of values and power, highlighting that safety has to include social harms and pluralistic desiderata. Varshney [98] put this plainly: *"Values should not be assumed universal. Contextual and relational values should come from the communities in which the LLM will be deployed"*. In addition, some of the literature explicitly addresses the divide between the fields, usually in the context of AI governance. Prunkl and Whittlestone [72] refer to *"two seemingly separate communities"*, arguing that the near- and long-term distinction *"risks causing confusion and fuelling conflict"* and that we should instead aim to *"establish new opportunities for collaboration and reduce adversarial dynamics"*. In a different example, Stix and Maas [90] propose that scholars *"converge and cooperate on selected mutually beneficial AI policy projects, while maintaining their distinct perspectives"*.

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